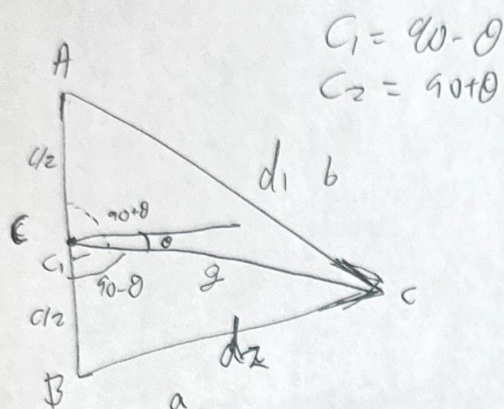




~~$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

$$b^2 = a^2 + c^2 - 2ac \cos(B)$$

$$\cos(B) = \frac{(c/2)^2 + (a)^2 - g^2}{2(c/2)a}$$

$$\cos(A) = \frac{(c/2)^2 + (b)^2 - g^2}{2(c/2)b}$$~~

α_1
 $\cos(C_1) = \frac{(c/2)^2 + g^2 - a^2}{2(c/2)(g)} = \cos(90^\circ - \theta) = \sin(\theta)$
 $\cos(C_2) = \frac{(c/2)^2 + g^2 - b^2}{2(c/2)(g)} = \cos(90^\circ + \theta) = -\sin(\theta)$
 α_2

$d = 1 \text{ meter}$
 $\frac{c}{2} = .1 \text{ meters}$
 $a = \frac{\sqrt{\frac{E_1}{E_2}} (.0005)(343)}{1 - \sqrt{\frac{E_1}{E_2}}}$
 $b = \frac{(.0005)(343)}{1 - \sqrt{E_1/E_2}}$
 $g = \sqrt{\frac{a^2 + b^2 - .0002}{2}}$
 $\theta = \sin^{-1} \left(\frac{.001 + g^2 - a^2}{2(.01)(g)} \right)$

$\alpha_1 = -\alpha_2$
 $(c/2)^2 + g^2 - a^2 = -(c/2)^2 - g^2 + b^2$
 $2(c/2)^2 + 2g^2 = b^2 + a^2$
 $g = \sqrt{\frac{b^2 + a^2 - 2(c/2)^2}{2}}$
 $\theta = \sin^{-1} \left(\frac{(c/2)^2 + g^2 - a^2}{2(c/2)(g)} \right)$
 $\Delta V = b - a$
 $b = \Delta t v + a$
 $\frac{a}{b} = \sqrt{k}$
 $a = \sqrt{k} b$
 $b = \Delta t(v) + \sqrt{k} b$
 $b(1 - \sqrt{k}) = \Delta t(v)$
 $a = \frac{\sqrt{k} \Delta t(v)}{1 - \sqrt{k}}$
 $b = \Delta t(v) / (1 - \sqrt{k})$